A Data Science Approach to AI-Powered Climate Change Insights: DS4Earth

DISSERTATION

Submitted in partial fulfillment of the requirements of the

Degree: MTech in Data Science and Engineering

By

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2022dc04349

Under the supervision of

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DSECLZG628T / **AIMLCZG628T DISSERTATION**

Dissertation Title: A Data Science Approach to AI-Powered Climate Change Insights: **DS4Earth**

Name of Supervisor : Gulab Chand Yadav

Name of Student : Rajesh Kumar Singh

ID No. of Student : 2022dc04349

Courses Relevant for the Project & Corresponding Semester:

S No	Subject Name-Short	Subject Name -Full	Semester
1	DWH	Data Warehousing	III
2	ML	Machine Learning	II
3	DVI	Data Visualization and Interpretation	II
4	DMML	Data Management for Machine Learning	II

Abstract

Climate change ranks among the most pressing challenges of our era, evidenced by rising global temperatures, erratic weather patterns, and surging greenhouse gas emissions. Tackling this crisis requires the ability to analyse vast, complex climate datasets using advanced data science. The DS4Earth project introduces an AI-powered climate analytics framework that merges data warehousing, machine learning, and interactive visualization to deliver actionable insights for research and policy.

The initiative begins with constructing a unified climate data warehouse, integrating heterogeneous datasets from authoritative sources like NASA, Copernicus, and the Indian Meteorological Department (IMD). Modelled on a star schema, the warehouse enables efficient OLAP (Online Analytical Processing), supporting deep spatial and historical analysis. An ETL (Extract, Transform, Load) pipeline streamlines the ingestion, cleansing, and consolidation of key parameters such as temperature anomalies, CO_2 concentration, rainfall, humidity, and sea-level rise.

After pre-processing steps—such as anomaly detection, normalization, and time-series transformation—Exploratory Data Analysis (EDA) uncovers hidden patterns and correlations using visual tools like heatmaps, line plots, and trend graphs. Machine learning models deploy for forecasting, while KMeans aids in climate zone classification. Regression analysis explores causal links, such as between CO_2 emissions and temperature shifts.

Insights are brought to life through dynamic dashboards and reports built with Tableau and Python, offering stakeholders clear, interactive views of climate trends, geographic risk zones, and future projections. These tools enhance communication and empower informed decision-making for climate resilience.

By combining AI with robust data warehousing and visual storytelling, DS4Earth boosts interpretability and drives evidence-based responses to climate change. Real-time API integration further broadens its utility, offering continuously updated insights for adaptive policymaking and scientific exploration.

In bridging the gap between raw climate data and actionable intelligence, DS4Earth showcases the transformative role of data-driven innovation in addressing the climate emergency.

Keywords:

Category	Keywords
Core Themes	Climate Change, Sustainability, Environmental Analytics
Data Engineering	Data Warehousing, Star Schema, ETL (Extract, Transform, Load), OLAP (Online Analytical Processing), Real-time API Integration
Data Sources	NASA, NOAA, CMIP6, Copernicus, Indian Meteorological Department (IMD)
Data Parameters	Temperature Anomalies, CO_2 Levels, Rainfall, Humidity, Sea-Level Rise
Machine Learning & AI	Exploratory Data Analysis (EDA), Time-Series Forecasting, LSTM, Random Forest, Clustering, Regression Analysis
Visualization & Tools	Interactive Dashboards, Trend Visualization Tableau, Python Visualization

II SEMESTER 24-25

DSECLZG628T / **AIMLCZG628T DISSERTATION**

Dissertation Outline (Abstract)

BITS ID No. **2022dc04349** **Name of Student:** **Rajesh Kumar Singh**

Name of Supervisor: Gulab Chand Yadav

Designation of Supervisor: Technical Architect, Hexaware Technologies

Qualification and Experience: M.Tech, 18+ years

Official E- mail ID of Supervisor: gulaby@hexaware.com

Topic of Dissertation: A Data Science Approach to AI-Powered Climate Change Insights: **DS4Earth**

Rajesh Kumar Singh

(Signature of Student)

Date: 22/05/2025

Gulab

(Signature of Supervisor)

Date: 22/05/2025

1.Broad Area of Work

This project lies at the intersection of Environmental Data Science, Artificial Intelligence (AI), Data Warehousing (DWH), and Data Visualization (DV), with a focus on applying Machine Learning (ML) to climate change analysis, forecasting, and decision support.

2.Objectives

The objectives of my project are as follows:

To unify and analyse climate data from trusted sources using a star schema-based data warehouse with ETL processes. To apply machine learning for forecasting and uncovering climate patterns and relationships. To deliver actionable insights through real-time API integration and interactive visualizations using Tableau and Python tools.

3.Scope of Work

Scope of this dissertation is to design and develop –

- The project involves designing a climate data warehouse and building ETL pipelines for integrating multi-source datasets.
- It includes applying machine learning for forecasting key climate indicators and detecting regional patterns.
- Developing interactive dashboards , real time API integration and enabling real-time insights for data-driven climate policy decisions.

4. Detailed Plan of Work

The plan of work should have tangible weekly or fortnightly milestones and deliverables, which can be measured to assess the adherence to the plan and therefore the rate of progress in the work. The plan of work can be specified in the table given below:

Serial Number of Task/Phases	**Tasks or subtasks to be done**	**Start Date-End Date**	Planned duration in weeks	Specific Deliverable in terms of the project
1	Preparation and Background Research - Dataset identification & sourcing - Requirement gathering & architecture design of data warehouse and ETL pipeline	07/05/2025 - 19/05/2025	2 weeks	Requirement specification and architecture design document

2	System Development - Data Warehouse Development - Data Pre-processing & EDA - Machine Learning Model Development - Real-Time API Integration - Visualization & Dashboard Development	20/05/2025 – 29/07/2025	11 weeks	Code
3	Testing and Evaluation - Functional Testing	30/07/2025 – 11/08/2025	2 weeks	Test Summary report
4	Documentation and Final Report	12/08/2025 – 18/08/2025	1 week	Final Project report

5. Literature References

The following are referred journals from the preliminary literature review.

S. No.	Title	Source / Publisher
1.	NASA GISTEMP	• NASA-GISS Surface Temperature Analysis
2.	India Meteorological Department	• India Meteorological Department
3.	Climate Change and Machine Learning: A Review (Rolnick et al., 2019)- Research Paper	• Climate Change with Machine Learning
4.	AI For Climate Change: 2024- Research Paper	• AI For Climate Change
5.	NOAA Climate Data Online (CDO)	• Climate Data Online
6.	IPCC AR6 Reports- Official Report	• The Intergovernmental Panel on Climate Change

Supervisor's Rating of the Technical Quality of this Dissertation Outline

EXCELLENT / GOOD / FAIR/ POOR (Please specify): EXCELLENT

Supervisor's suggestions and remarks about the outline (if applicable).

Date 22/05/2025

Gulab

(Signature of Supervisor)

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